

Project Title:

Customer Churn Analysis and Prediction using Data Analytics

**Prepared by:
Rramandeip Singh**

Data Analytics

Humber College

Dataset used: Telco Customer Churn Dataset- www.kaggle.com/datasets/blatchar/telco-customer-churn

1. Problem Statement

Customer churn is a major challenge for telecom companies because losing existing customers reduces recurring revenue and increases the cost of acquiring new customers. In highly competitive markets, retaining customers is often more cost-effective than attracting new ones.

The objective of this project is to analyze customer data to identify the key factors influencing churn and to develop a predictive model that can classify customers who are likely to leave the company. This project combines descriptive analytics, exploratory data analysis, and predictive analytics to support better customer retention strategies.

2. Dataset Description

The dataset used for this project is the Telco Customer Churn dataset, which contains customer account and service information from a telecommunications company.

The dataset includes customer demographics (gender, senior status, partner, dependents), account information (tenure, contract type, paperless billing, payment method), billing variables (monthly and total charges), and service features (phone service, multiple lines, internet service, security, tech support, streaming). The target variable is churn, indicating whether a customer has left (Yes) or stayed (No).

This dataset is highly suitable for customer retention analysis because it combines behavioral, billing, and service usage variables.

3. Dataset Analysis and Observations

The dataset was analyzed using **univariate**, **bivariate**, and **multivariate** techniques to identify churn patterns. The overall churn rate is ~26.5%, with many customers on mid-to-high pricing plans and relatively short tenure. Churn is highest among month-to-month contract users and customers in their first 12 months, highlighting the importance of early retention and contract commitment. Higher churn is also observed among electronic check users and fiber optic customers. Overall, churn is driven by a combination of low tenure, weaker commitment, higher charges, and payment behavior.

Key Findings

Contract type is a major predictor of churn.

New customers are the most likely to leave.

Higher monthly charges increase churn likelihood.

Electronic check users are a vulnerable segment.

Churn risk can be effectively predicted using customer data.

4. Proposed Analytical/Prediction Model

Section	Details
Model Choice	Logistic Regression for binary churn prediction; efficient and interpretable
Inputs	Tenure, charges, contract type, payment method, services, demographics
Preprocessing	One-hot encoding (categorical), standardization (numerical)
Process	80/20 train-test split; model trained and evaluated

Output:

The model predicts whether a customer is likely to churn and estimates churn probability. This can help the company prioritize retention efforts toward high-risk customers.

5. Results and Discussions

Metric	Result	Insight
Accuracy	75.23%	Correctly classifies ~3 out of 4 customers overall
Precision	52.16%	Moderate accuracy in churn predictions; some false positives present
Recall	80.75%	Effectively identifies most actual churn cases (strong detection rate)
F1 Score	63.38%	Balanced performance between precision and recall

Why High Recall Matters Most

For churn prediction, recall is more important than precision because the business priority is identifying as many at-risk customers as possible before they leave. It is often acceptable to target some customers who may not churn if it means successfully retaining real churn-risk customers.

Actionable business insights:

Focus Area	Recommendation
Early Retention	Provide onboarding support and incentives to customers in their first year
Contract Upgrades	Encourage month-to-month users to switch to long-term contracts
High-Value Customer Support	Offer enhanced support to customers with higher monthly charges
Predictive Campaigns	Use churn scores to proactively target and retain high-risk customers

In conclusion

This project demonstrates how data analytics can be used to understand and reduce customer churn. The analysis showed that contract type, tenure, payment method, and monthly charges strongly influence churn behavior. The Logistic Regression model performed effectively, particularly with high recall, making it valuable for real-world churn prevention strategies. By using these insights, telecom companies can reduce churn, improve customer satisfaction, and increase long-term profitability.